

```
for i in {1..3}; do
    echo var$i
```

6406534158571. ✖ done

6406534158572. ✔ echo \$var{1..3}

Sub-Section Number : 5
Sub-Section Id : 640653189713
Question Shuffling Allowed : Yes

Question Number : 302 Question Id : 6406531230829 Question Type : SA

Correct Marks : 6

Question Label : Short Answer Question

The value of the BASH_SUBSHELL after executing the following command
\$ (ls;(date;(file *|echo \$BASH_SUBSHELL)))

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 303 Question Id : 6406531230830 Question Type : SA

Correct Marks : 6

Question Label : Short Answer Question

What is the output of the following command
sh -c 'exit 405'; echo \$?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

149

Section Id :	64065386918
Section Number :	14
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	24
Number of Questions to be attempted :	24
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653189714
Question Shuffling Allowed :	No

Question Number : 304 Question Id : 6406531230831 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : PROGRAMMING CONCEPTS USING JAVA (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE [TOP](#) FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534158575.  YES

6406534158576.  NO

Sub-Section Number :	2
Sub-Section Id :	640653189715
Question Shuffling Allowed :	Yes

Question Number : 305 Question Id : 6406531230832 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the code given below.

```
class Shared {
    static int count = 0;
}

class Adder extends Thread {
    public void run() {
        for(int i=0; i<1000; i++) {
            Shared.count++;
        }
    }
}

public class Test {
    public static void main(String[] args) throws InterruptedException {
        Thread t1 = new Adder();
        Thread t2 = new Adder();
        t1.start();
        t2.start();
        t1.join();
        t2.join();
        System.out.print(Shared.count);
    }
}
```

Choose the correct option.

Options :

- 6406534158577. ✖ The code always prints 2000.
- 6406534158578. ✔ The code prints any number in the range 1000 to 2000
- 6406534158579. ✖ The code prints any number in the range 0 to 1000
- 6406534158580. ✖ The code always prints 1000.

Question Number : 306 Question Id : 6406531230833 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following Java code that uses chained exceptions.

```
class FileStorageException extends Exception {
    public FileStorageException(String message) {
        super(message);
    }
}

class FileUploadException extends Exception {
    public FileUploadException(String message) {
        super(message);
    }
}

class FileService {
    public void uploadFile(String filename) throws FileUploadException {
        try {
            // if(do not have enough disk space to store the file)
            throw new FileStorageException("Insufficient disk space");
            // else upload the file
        } catch (FileStorageException e) {
            FileUploadException fue = new FileUploadException("Failed to upload");
            fue.initCause(e);
            throw fue;
        }
    }
}

public class FileApp {
    public static void main(String[] args) {
        FileService fileService = new FileService();
        try {
            fileService.uploadFile("report.pdf");
        } catch (FileUploadException e) {
            System.out.println(e.getMessage());
            System.out.println("Reason: " + e.getCause().getMessage());
        }
    }
}
```

Which of these could be the output of this code?

Options :

Failed to upload
6406534158581. ✓ Reason: Insufficient disk space

Insufficient disk space
6406534158582. ✗ Reason: Failed to upload

Reason: Insufficient disk space
6406534158583. ✗

6406534158584. ✖ Reason: Failed to upload

Question Number : 307 Question Id : 6406531230834 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the two Java files given below.

Animal.java:

```
package creatures;
public class Animal {
    private String name;
    private String habitat;
    private int legs;
    // Constructor to initialize instance variables
    public String getName(String name) {
        return name;
    }
    protected String getHabitat(String habitat) {
        return habitat;
    }
    private int getLegs(int legs) {
        return legs;
    }
    public int showLegs(int legs) {
        return getLegs(legs);
    }
}
```

Test.java:

```
package Animals;
public class Test {
    public static void main(String[] args) {
        Animal myAnimal = new Animal();
        System.out.println("Number of legs: " + myAnimal.showLegs(4)); // LINE 1
        System.out.println(myAnimal.getHabitat("Forest")); // LINE 2
        System.out.println(myAnimal.getLegs(4)); // LINE 3
    }
}
```

Choose the correct option.

Options :

6406534158585. ✖ LINE 1 and LINE 2 generate compilation errors.

6406534158586. ✔ LINE 2 and LINE 3 generate compilation errors.

6406534158587. ✖ Only LINE 3 generates compilation error.

6406534158588. ✖ Only LINE 2 generates compilation error.

Question Number : 308 Question Id : 6406531230835 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following Java code.

```
class CreditCard {
    private double limit, balance, interestRate;
    public CreditCard(double l, double b, double i) {
        assert i > 0 : "Interest rate must be positive";
        limit = l;
        balance = b;
        interestRate = i;
    }
    private boolean isValidTransaction(double amount) {
        return (balance + amount <= limit);
    }
    public double processPayment(double amount) {
        assert amount > 0 : "Payment must be positive";
        assert isValidTransaction(amount) : "Transaction exceeds credit limit";

        balance += amount;
        return balance;
    }
}

class CreditCardTest {
    public static void main(String[] args) {
        CreditCard card = new CreditCard(5000.00, 4000.00, 18.99);
        System.out.println(card.processPayment(2000.00));
    }
}
```

Choose the correct option when the class is executed as:

java -ea CreditCardTest

Options :

6406534158589. ✖ This code generates output: 6000

6406534158590. ✖ This code generates AssertionError: Payment must be positive

6406534158591. ✔ This code generates AssertionError: Transaction exceeds credit limit

6406534158592. ✖ This code generates AssertionError: Interest rate must be positive

Question Number : 309 Question Id : 6406531230836 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Java code given below.

```
class Calculator {  
    public <T extends Number> void calculate(T value) { // LINE 1  
        System.out.println("Calculating: " + (value.doubleValue() * 2));  
    }  
    public <U extends Number> void calculate(U value) { // LINE 2  
        System.out.println("Double Calculation: " + (value.doubleValue() * 4));  
    }  
    public <T> void calculate(T value) { // LINE 3  
        System.out.println("Printing: " + value);  
    }  
}  
public class Test {  
    public static void main(String[] args) {  
        Calculator calc = new Calculator();  
        calc.calculate(5);  
        calc.calculate(2.1);  
        calc.calculate("Hello");  
    }  
}
```

Choose the correct option.

Options :

6406534158593. ✖ LINE 2 and LINE 3 generates a compilation error because method calculate() is already defined with the same signature.

6406534158594. ✖

LINE 1 and LINE 3 generates a compilation error because method `calculate()` is already defined with the same signature.

6406534158595. ✓ Only LINE 2 generates a compilation error because method `calculate()` is already defined with the same signature.

This program generates output:
Calculating: 10
Double Calculation: 8.4
Hello

6406534158596. ✗

Question Number : 310 Question Id : 6406531230837 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Java code given below.

```
class Executive {
    String name;
    public Executive(String n) {
        this.name = n;
    }
    public String toString() {
        return "name = " + name;
    }
}
class Director extends Executive {
    int project_count;
    public Director(String n, int p) {
        super(n);
        project_count = p;
    }
    public Director(Director d) {
        super(d.name);
        project_count = d.project_count;
    }
    public String toString() {
        return super.toString() + ", project_count = " + project_count;
    }
}

public class Corporation {
    public static void main(String args[]) {
        Executive d1 = new Director("Anjali", 5);
        Executive d2 = new Director((Director) d1);
        d2.name = "Rajesh";
        System.out.println(d1 + "\n" + d2);
    }
}
```

What will the output be?

Options :

name = Rajesh

6406534158597. ✖ name = Rajesh

name = Rajesh, project_count = 5

6406534158598. ✖ name = Rajesh, project_count = 5

name = Anjali, project_count = 5

6406534158599. ✔ name = Rajesh, project_count = 5

```
        name = Anjali
6406534158600. ✖ name = Rajesh
```

Question Number : 311 Question Id : 6406531230838 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Java code given below.

```
abstract class Vehicle {
    abstract void start();
    void fuel() { // LINE 1
        System.out.println("Fueling vehicle");
    }
}
class Car extends Vehicle {
    void start() {
        System.out.println("Starting car");
    }
    void fuel() {
        System.out.println("Filling petrol in car");
    }
}
class Bike extends Vehicle {
    void start() {
        System.out.println("Starting bike");
    }
    void fuel() {
        System.out.println("Filling petrol in bike");
    }
}
public class Test {
    public static void main(String[] args) {
        Vehicle v1 = new Car(); // LINE 2
        Vehicle v2 = new Bike(); // LINE 3
        v1.fuel();
        v1.start();
        v2.fuel();
        v2.start();
    }
}
```

Choose the correct option.

Options :

LINE 1 generates compilation error because abstract class must contain only abstract methods.

6406534158601. ✖

6406534158602. ✖ LINE 2 and LINE 3 generate compilation errors because a reference variable of type Vehicle cannot store the objects of type Car and Bike.

This program generates the output:

Filling petrol in car
Starting car
Filling petrol in bike
Starting bike

6406534158603. ✔

This program generates the output:

Fueling vehicle
Starting car
Fueling vehicle
Starting bike

6406534158604. ✖

Question Number : 312 Question Id : 6406531230839 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Java code given below.

```
interface LightControl {
    void turnOn(); //LINE 1
    abstract void turnOff();
}
interface TemperatureControl {
    abstract void setTemperature(int temp);
}
class SmartHome implements LightControl, TemperatureControl { //LINE 2
    public void turnOn() {
        System.out.println("Lights turned ON");
    }

    public void turnOff() {
        System.out.println("Lights turned OFF");
    }

    public void setTemperature(int temp) {
        System.out.println("Temperature set to " + temp + "°C");
    }
}
public class Test {
    public static void main(String[] args) {
        LightControl home = new SmartHome();
        home.turnOn();
        home.turnOff();
        home.setTemperature(22); //LINE 3
    }
}
```

What will the output be?

Options :

6406534158605. ✖ LINE 1 generates compilation error `turnOn()` is not declared as abstract
6406534158606. ✖ LINE 2 generates compilation error as a class cannot implement multiple interfaces
6406534158607. ✖ This program generates the output:
Lights turned ON
Lights turned OFF
Temperature set to 22°C
6406534158608. ✔ LINE 3 generates compilation error as `setTemperature()` method is not accessible

Question Number : 313 Question Id : 6406531230840 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Java code given below.

```
import java.util.*;
public class Test {
    public static void main(String[] args) {
        Map<String, Integer> exam1 = new TreeMap<String, Integer>();
        exam1.put("Amit", 85);
        exam1.put("Neha", 78);
        exam1.put("Rahul", 90);
        exam1.put("Priya", 88);

        Map<String, Integer> exam2 = new TreeMap<String, Integer>();
        exam2.put("Amit", 80);
        exam2.put("Neha", 85);
        exam2.put("Rahul", 92);
        exam2.put("Priya", 86);

        Map<String, Integer> totalMarks = new TreeMap<String, Integer>();

        for (Map.Entry<String, Integer> e1 : exam1.entrySet())
            totalMarks.put(e1.getKey(), e1.getValue());

        for (Map.Entry<String, Integer> e2 : exam2.entrySet())
            totalMarks.merge(e2.getKey(), e2.getValue(), Integer::sum); // LINE 1

        System.out.println(totalMarks);
    }
}
```

Choose the correct option.

Options :

6406534158609. ✖ LINE 1 generates a compilation error because of an invalid key.

The program generates the output:

6406534158610. ✔ {Amit=165, Neha=163, Priya=174, Rahul=182}

The program generates the output:

6406534158611. ✖ {Amit=165, Neha=163, Rahul=182, Priya=174}

The program generates the output:

6406534158612. ✖ {Rahul=182, Priya=174, Neha=163, Amit=165}

Question Number : 314 Question Id : 6406531230841 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Method `Stream.iterate(e, f)` returns an infinite sequential ordered `Stream` produced by iterative application of a function `f` to an initial element `e`, producing a `Stream` consisting of `e`, `f(e)`, `f(f(e))`, etc. Based on the above information, consider the code given below, and answer the question that follows.

```
import java.util.stream.Stream;
public class Test {
    public static void main(String[] args) {
        Stream.iterate(2, n -> n + 3)
            .map(n -> n * 2)
            .filter(n -> n % 5 == 0)
            .limit(3)
            .forEach((x) -> System.out.print(x + " "));
    }
}
```

What will the output be?

Options :

6406534158613. ✖ 4 10 16

6406534158614. ✔ 10 40 70

6406534158615. ✖ 5 20 35

6406534158616. ✖ 20 50 80

Question Number : 315 Question Id : 6406531230842 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the code given below.

```
class Rider implements Cloneable {
    String name;
    int[] distances;
    public Rider(String n, int[] d) {
        name = n;
        distances = d;
    }
    public Object clone() throws CloneNotSupportedException {
        return super.clone();
    }
}

public class Test{
    public static void main(String[] args) throws CloneNotSupportedException {
        int[] d = {50, 60, 70};
        Rider r1 = new Rider("Raj", d);
        Rider r2 = (Rider) r1.clone();
        Rider r3 = r1;
        r2.distances[1] = 90;
        r3.name = "Amit";

        System.out.println(r1.name + " " + r1.distances[1]);
        System.out.println(r2.name + " " + r2.distances[1]);
        System.out.println(r3.name + " " + r3.distances[1]);
    }
}
```

What will the output be?

Options :

Amit 90

Raj 90

6406534158617. ✓ Amit 90

Amit 60

Raj 90

Amit 60

6406534158618. ✗

Raj 60

Raj 90

Amit 60

6406534158619. ✗

Raj 60

Raj 60

Amit 60

6406534158620. ✗

Question Number : 316 Question Id : 6406531230843 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Method `Optional.ofNullable(T value)` returns an `Optional` describing the specified value, if non-null, otherwise returns an empty `Optional`. Based on this description, consider the code given below, and answer the question that follows.

```
import java.util.*;
class GrindingMachine {
    String brand;
    String powerSource;
    public GrindingMachine(String b, String pS) {
        this.brand = b;
        this.powerSource = pS;
    }
}
public class Test {
    public static void main(String[] args) {
        var machineList = new ArrayList<GrindingMachine>();
        machineList.add(new GrindingMachine("Bosch", "Electric"));
        machineList.add(new GrindingMachine("Hitachi", null));
        for (GrindingMachine obj : machineList) {
            Optional<String> op1 = Optional.ofNullable(obj.powerSource);
            op1.ifPresent(source -> System.out.println(source));
        }
    }
}
```

Choose the correct option.

Options :

This program generates the output:

Electric

6406534158621. ✖ null

This program generates the output:

Electric

6406534158622. ✔

This program terminates abnormally due to `NullPointerException` without printing any value.

6406534158623. ✖

6406534158624. ✖

This program terminates abnormally due to NullPointerException after printing the value:

Electric

Question Number : 317 Question Id : 6406531230844 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the code given below.

```
import java.io.*;

class Laptop implements Serializable {
    private double price;
    private transient String model;
    private transient int serialNum;
    // Constructor to initialize instance variables
    public String toString() {
        return "price=" + price + ", model=" + model + ", serialNum=" + serialNum;
    }

    private void writeObject(ObjectOutputStream out) throws IOException {
        out.defaultWriteObject();
        out.writeInt(serialNum + 100);
    }

    private void readObject(ObjectInputStream in) throws Exception {
        in.defaultReadObject();
        serialNum = in.readInt() - 100;
    }
}

public class Test {
    public static void main(String[] args) throws Exception {
        var fos = new FileOutputStream("Laptop.txt");
        var oos = new ObjectOutputStream(fos);
        Laptop l1 = new Laptop(899.99, "Dell", 987654);
        oos.writeObject(l1);

        var fis = new FileInputStream("Laptop.txt");
        var ois = new ObjectInputStream(fis);
        Laptop obj = (Laptop) ois.readObject();
        System.out.println(obj);
    }
}
```

What will the output be?

Options :

6406534158625. ✖ price=899.99, model=Dell, serialNum=0
6406534158626. ✖ price=899.99, model=Dell, serialNum=987654
6406534158627. ✔ price=899.99, model=null, serialNum=987654
6406534158628. ✖ price=899.99, model=null, serialNum=0

Sub-Section Number : 3
Sub-Section Id : 640653189716
Question Shuffling Allowed : Yes

Question Number : 318 Question Id : 6406531230845 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider the Java code given below.

Assume the file "data.txt" already contains the following text:

Welcome to Java Programming

```
import java.io.*;
public class Test {
    public static void main(String[] args) {
        try {
            FileInputStream fis = new FileInputStream("data.txt");
            int i;
            while ((i = fis.read()) != -1) {
                if (i == 'a') {
                    System.out.print('*');
                } else {
                    System.out.print((char) i);
                }
            }
            fis.close();
        } catch (IOException e) {
            System.out.println("File not found!");
        }
    }
}
```

Choose the correct option regarding the output of the program:

Options :

6406534158629. ✓ Welcome to J*v* Progr*mming

6406534158630. ✖ Welcome to Java Programming

6406534158631. ✖ Welcome to J*va Programming

6406534158632. ✖ File not found!

Question Number : 319 Question Id : 6406531230846 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Method `Collectors.partitioningBy(predicate)` returns a `Collector` which partitions the input elements according to a predicate, and organizes them into a `Map<Boolean, List<T>>`.

```
import java.util.*;
import java.util.stream.Collectors;
public class Test {
    public static void main(String[] args) {
        var numbers = new ArrayList<Integer>(); // LINE 1

        for (int i = 10; i <= 30; i += 5) {
            numbers.add(i);
        }

        Map<Boolean, List<Integer>> partitioned = numbers.stream()
            .collect(Collectors.partitioningBy(num -> num % 2 == 0));

        System.out.println("true -> " + partitioned.get(true));
        System.out.println("false -> " + partitioned.get(false));
    }
}
```

Choose the correct option.

Options :

This program generates the output:

true -> [10, 20, 30]

6406534158633. ✓ false -> [15, 25]

6406534158634. ✖

This program generates the output:

true -> [15, 25]

false -> [10, 20, 30]

6406534158635. ✖ This program generates compile time error at LINE 1 because the reference type of list is not specified.

6406534158636. ✖ This program generates a runtime error due to incompatible stream elements.

Sub-Section Number : 4

Sub-Section Id : 640653189717

Question Shuffling Allowed : Yes

Question Number : 320 Question Id : 6406531230847 Question Type : MCQ

Correct Marks : 6

Question Label : Multiple Choice Question

Consider the Java code given below.

```
import javax.swing.*;
import java.awt.*;

public class FeedbackForm extends JFrame {
    JPanel pnlName, pnlFeedback, pnlSubmit;
    JLabel lblName, lblFeedback;
    JTextField txtName;
    JTextArea txtFeedback;
    JButton btnSubmit;

    public FeedbackForm() {
        lblName = new JLabel("Name:");
        lblFeedback = new JLabel("Feedback:");
        txtName = new JTextField(15);
        txtFeedback = new JTextArea(5, 20);
        btnSubmit = new JButton("Submit");

        pnlName = new JPanel();
        //add lblName and txtName to pnlName
        pnlFeedback = new JPanel();
        //add lblFeedback and txtFeedback to pnlFeedback
        pnlSubmit = new JPanel();
        //add btnSubmit to pnlSubmit

        //CODE BLOCK

        setVisible(true);
        setSize(400, 300);
    }

    public static void main(String[] args) {
        new FeedbackForm();
    }
}
```

Choose the correct option to be filled in place of CODE BLOCK such that the above program produces the GUI given below.

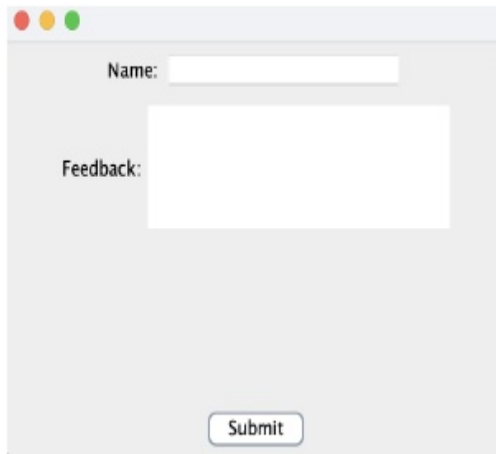


Figure 1: Feedback Form

Options :

6406534158637. ✖ `add(pnlName, "Center");`
`add(pnlFeedback, "North");`
`add(pnlSubmit, "South");`

6406534158638. ✖ `add(pnlName, "North");`
`add(pnlFeedback, "South");`
`add(pnlSubmit, "Center");`

6406534158639. ✔ `add(pnlName, "North");`
`add(pnlFeedback, "Center");`
`add(pnlSubmit, "South");`

6406534158640. ✖ `add(pnlName, "South");`
`add(pnlFeedback, "Center");`
`add(pnlSubmit, "North");`

Sub-Section Number :

5

Sub-Section Id :

640653189718

Question Shuffling Allowed :

Yes

Question Number : 321 Question Id : 6406531230848 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements is/are correct?

Options :

6406534158641. ✔ The heap stores dynamically allocated data.

6406534158642. ✖ Activation records created for functions are allocated on the heap.

6406534158643. ✔ The heap storage needs to be explicitly requested by the programmer.

6406534158644. ✖ Storage allocated on the heap cannot be deallocated in any way.

Question Number : 322 Question Id : 6406531230849 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the Java code given below.

```
import java.util.concurrent.ConcurrentHashMap;
class UpdaterThread extends Thread {
    private ConcurrentHashMap<String, Integer> map;
    public UpdaterThread(ConcurrentHashMap<String, Integer> m) {
        map = m;
    }
    public void run() {
        map.put("C", 3);
        map.put("D", 4);
    }
}
class IteratorThread extends Thread {
    private ConcurrentHashMap<String, Integer> map;
    public IteratorThread(ConcurrentHashMap<String, Integer> m) {
        map = m;
    }
    public void run() {
        for (String key : map.keySet()) {
            System.out.println(key + ": " + map.get(key));
        }
    }
}
public class Test {
    public static void main(String[] args) {
        ConcurrentHashMap<String, Integer> map = new ConcurrentHashMap<>();
        map.put("A", 1);
        map.put("B", 2);
        UpdaterThread updaterThread = new UpdaterThread(map);
        IteratorThread iteratorThread = new IteratorThread(map);
        updaterThread.start();
        iteratorThread.start();
        try {
            updaterThread.join();
            iteratorThread.join();
        } catch (InterruptedException e) {
            e.printStackTrace();
        }
        System.out.println("Final Map: " + map);
    }
}
```

Which of the following is/are true about the given code.

Options :

This program may generate the output:

A: 1

B: 2

C: 3

D: 4

Final Map: {A=1, B=2, C=3, D=4}

6406534158645. ✓

The program may generate the output:

C: 3

D: 4

Final Map: {A=1, B=2, C=3, D=4}

6406534158646. ✖

The program may generate the output:

A: 1

B: 2

C: 3

Final Map: {A=1, B=2, C=3, D=4}

6406534158647. ✓

The program always generates the output:

B: 2

A: 1

Final Map: {A=1, B=2}

6406534158648. ✖

Question Number : 323 Question Id : 6406531230850 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the code given below.

```
class Library {
    private boolean isBookAvailable = true;
    public synchronized void borrowBook(String readerName) {
        if (isBookAvailable) {
            System.out.println(readerName + " borrowed the book.");
            isBookAvailable = false;
        } else {
            System.out.println(readerName + " could not borrow the book.");
        }
    }
}

class Reader implements Runnable {
    private Library library;
    private String readerName;
    public Reader(Library lib, String rn) {
        this.library = lib;
        this.readerName = rn;
    }
    public void run() {
        library.borrowBook(readerName);
    }
}

public class Test {
    public static void main(String[] args) {
        Library library = new Library();
        Thread reader1 = new Thread(new Reader(library, "Rahul"));
        Thread reader2 = new Thread(new Reader(library, "Ram"));
        Thread reader3 = new Thread(new Reader(library, "Raghu"));
        reader1.start();
        reader2.start();
        reader3.start();
    }
}
```

Which of the following options is/are possible result/s of the above code?

Options :

This code generates the output:

Rahul borrowed the book.

Raghu could not borrow the book.

Ram could not borrow the book.

6406534158649. ✓

6406534158650. ✗

This code generates the output:

Rahul borrowed the book.

Ram borrowed the book.

Raghu borrowed the book.

This code generates the output:

Raghu borrowed the book.

Rahul could not borrow the book.

6406534158651. ✓

Ram could not borrow the book.

This code generates the output:

Rahul borrowed the book.

Ram borrowed the book.

6406534158652. ✗

Raghu could not borrow the book.

Question Number : 324 Question Id : 6406531230851 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the Java code given below that processes `Drawable` objects. From among the options, identify the appropriate function header for function `drawAll` that takes as input a collection of `Drawable` objects and calls the `draw` method on each.

```
import java.util.*;
interface Drawable {
    void draw();
}
class Rectangle implements Drawable {
    // Constructor
    // method draw() that prints rectangle details
}
class Triangle implements Drawable {
    // Constructor
    // method draw() that prints triangle details
}
public class Canvas {
    // LINE 1: FUNCTION HEADER
    {
        // invokes draw() on each element
    }
    public static void main(String[] args) {
        Set<Drawable> shapes = new HashSet<>();
        shapes.add(new Rectangle(4, 5));
        shapes.add(new Triangle(3, 4, 5));
        drawAll(shapes);
    }
}
```

Choose the correct option(s).

Options :

6406534158653. ✖ `public static void drawAll(Collection<?> elements)`

6406534158654. ✖ `public static <T> void drawAll(Collection<T> elements)`

6406534158655. ✔ `public static <T extends Drawable> void drawAll(Collection<T> elements)`

6406534158656. ✔ `public static void drawAll(Set<Drawable> elements)`

Question Number : 325 Question Id : 6406531230852 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the code given below.

```
class Person {
    public void work() {
        System.out.println("works as a person.");
    }
}
class Teacher extends Person {
    public void work() {
        System.out.println("Works as teacher.");
    }
    public void grade() {
        System.out.println("Grading papers.");
    }
}
class MathTeacher extends Teacher {
    public void solveEquation() {
        System.out.println("Solving complex equations.");
    }
}
public class SchoolTest {
    public static void main(String[] args) {
        Person p = new MathTeacher(); // LINE 1
        p.work();
        p.grade(); // LINE 2
        p.solveEquation(); // LINE 3
    }
}
```

Choose the correct option(s).

Options :

This program generates the output:

Works as teacher.

Grading papers.

6406534158657. ✖ Solving complex equations.

LINE 1 generates a compilation error because a variable of type Person cannot refer to an object of type MathTeacher.

6406534158658. ✖

LINE 2 generates a compilation error because the method grade() is not defined in the Person class.

6406534158659. ✔

6406534158660. ✔

LINE 3 generates a compilation error because the method solveEquation() is not defined in the Person class.

Sub-Section Number : 6
Sub-Section Id : 640653189719
Question Shuffling Allowed : Yes

Question Number : 326 Question Id : 6406531230853 Question Type : MSQ

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the following Java code.

```
1. class Animal {  
2.     protected final void makeSound() {  
3.         System.out.println("Animal makes sound");  
4.     }  
5. }  
6. class Bird {  
7.     public void fly() {  
8.         System.out.println("Bird flies");  
9.     }  
10. }  
11. class Parrot extends Animal, Bird {  
12.     public void makeSound() {  
13.         System.out.println("Parrot squawks");  
14.     }  
15. }  
16. public class TestProgram {  
17.     public static void main(String[] args) {  
18.         Parrot p1 = new Animal();  
19.     }  
20. }
```

Identify the line/s which has/have errors.

Options :

- 6406534158661. ✖ Line 2
- 6406534158662. ✖ Line 7
- 6406534158663. ✔ Line 11
- 6406534158664. ✔ Line 12
- 6406534158665. ✔ Line 18

Question Number : 327 Question Id : 6406531230854 Question Type : MSQ

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the Java code given below.

```
import java.awt.*;
import javax.swing.*;
import java.awt.event.*;

class OrderDemo extends JFrame implements ActionListener {
    JButton btnOrder, btnCancel;
    JLabel label;
    JPanel p1, p2;

    public OrderDemo() {
        p1 = new JPanel();
        p2 = new JPanel();

        btnOrder = new JButton("Order");
        btnCancel = new JButton("Cancel Order");
        label = new JLabel("Select an option");

        p1.add(btnOrder);
        p1.add(btnCancel);
        p2.add(label);

        add(p1, "Center");
        add(p2, "South");

        btnOrder.setActionCommand("Order");
        btnCancel.setActionCommand("Cancel");
        btnOrder.addActionListener(this);
        btnCancel.addActionListener(this);

        setSize(350, 150);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        // CODE SEGMENT
    }
}

public class OrderTest {
    public static void main(String[] args) {
        new OrderDemo();
    }
}
```

Choose the correct code segment to be filled inside the method `actionPerformed()` such that on clicking the Order button, the label text changes to Order placed successfully, and on clicking the Cancel Order button, the label text changes to Order canceled.



Options :

```
if (e.getActionCommand().equals("Order"))
    label.setText("Order placed successfully");
else if (e.getActionCommand().equals("Cancel"))
    label.setText("Order canceled");
```

6406534158666. ✓

```
if (e.getSource().equals("Order"))
    label.setText("Order placed successfully");
else if (e.getSource().equals("Cancel"))
    label.setText("Order canceled");
```

6406534158667. ✗

```
if (e.getSource().equals(btnOrder))
    label.setText("Order placed successfully");
else if (e.getSource().equals(btnCancel))
    label.setText("Order canceled");
```

6406534158668. ✓

```
if (e.getActionCommand().equals(Order))
    label.setText("Order placed successfully");
else if (e.getActionCommand().equals(Cancel))
    label.setText("Order canceled");
```

6406534158669. ✗

TDS

Section Id :	64065386919
Section Number :	15
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	17
Number of Questions to be attempted :	17

Section Marks :	40
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653189720
Question Shuffling Allowed :	No

Question Number : 328 Question Id : 6406531230855 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : TOOLS IN DATA SCIENCE (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534158670. ✓ YES

6406534158671. ✗ NO

Sub-Section Number :	2
Sub-Section Id :	640653189721
Question Shuffling Allowed :	Yes

Question Number : 329 Question Id : 6406531230856 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

You receive a tab-delimited file where some numeric fields are interpreted as dates (e.g., "3/10" becomes "March 10" in Excel). What is the most effective way to handle this in Text to-Columns?

Options :

6406534158672. ✗ Change the delimiter to commas before opening the file

6406534158673. ✓ Format the column as Text before applying Text-to-Columns

6406534158674. ✗ Use the General data format option in Text-to-Columns

6406534158675. ✗ Manually change the affected cells back to numbers after splitting

Question Number : 330 Question Id : 6406531230858 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In Python, which statement correctly handles exceptions and ensures that cleanup code is executed?

Options :

6406534158680. ✖ `try-except`

6406534158681. ✔ `try-except-finally`

6406534158682. ✖ `if-else-finally`

6406534158683. ✖ `try-catch-finally`

Question Number : 331 Question Id : 6406531230859 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A research team has a pandas dataframe of medical trial participants. To quickly identify how many different treatment groups exist, which Pandas method would be most efficient?

Options :

6406534158684. ✖ ``len(set(column))``

6406534158685. ✖ ``column.count()``

6406534158686. ✔ ``column.nunique()``

6406534158687. ✖ ``column.value_counts()``

Question Number : 332 Question Id : 6406531230860 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In Git, what is the purpose of the `.gitignore` file?

Options :

6406534158688. ✖ To track changes to specific files

6406534158689. ✔ To specify files that Git should not track

6406534158690. ✖ To define Git repository settings

6406534158691. ✖ To list contributors to the repository

Question Number : 333 Question Id : 6406531230861 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A developer wants to examine API request headers and responses to debug an issue with a failing API call. Where in Chrome DevTools should they look?

Options :

6406534158692. ✖ Console tab

6406534158693. ✖ Application tab

6406534158694. ✔ Network tab

6406534158695. ✖ Sources tab

Question Number : 334 Question Id : 6406531230863 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A pharmaceutical researcher is testing a new drug's effectiveness. If the p-value is 0.03 in a regression analysis, what can be concluded about the drug's impact?

Options :

6406534158700. ✖ The drug has no significant effect

6406534158701. ✔ The result is statistically significant at a 5% significance level

6406534158702. ✖ More data collection is needed

6406534158703. ✖ The sample size is too small

Question Number : 335 Question Id : 6406531230864 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A data analyst receives a messy dataset with inconsistent company names like "Apple", "Apple Inc.", and "Apple Incorporated". Which tool would best standardize these entries?

Options :

6406534158704. ✖ Excel's Find and Replace

6406534158705. ✔ OpenRefine's clustering feature

6406534158706. ✖ Python's string manipulation

6406534158707. ✖ Manual editing

Question Number : 336 Question Id : 6406531230865 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A web developer wants to analyze the load time and resources of a webpage. Which Chrome DevTools feature provides comprehensive network request information?

Options :

6406534158708. ✖ Console tab

6406534158709. ✔ Network tab

6406534158710. ✖ Elements tab

6406534158711. ✖ Sources tab

Question Number : 337 Question Id : 6406531230866 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following is NOT a feature typically found in modern IDEs?

Options :

6406534158712. ✖ Syntax highlighting

6406534158713. ✖ Integrated debugger

6406534158714. ✖ Version control integration

6406534158715. ✔ Native hardware virtualization

Question Number : 338 Question Id : 6406531230867 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following describes the primary use of Kumu?

Options :

6406534158716. ✖ Building financial models

6406534158717. ✔ Creating system maps and visualizing relationships

6406534158718. ✖ Performing real-time data analysis

6406534158719. ✖ Designing geographic maps for navigation

Sub-Section Number : 3

Sub-Section Id : 640653189722

Question Shuffling Allowed : Yes

Question Number : 339 Question Id : 6406531230857 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which HTTP method is idempotent, meaning multiple identical requests have the same effect as a single request?

Options :

6406534158676. ✖ POST

6406534158677. ✔ PUT

6406534158678. ✖ GET

6406534158679. ✖ DELETE

Sub-Section Number : 4

Sub-Section Id : 640653189723

Question Shuffling Allowed : Yes

Question Number : 340 Question Id : 6406531230862 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

A researcher needs to perform spatial data operations like buffering and visualize the results on an interactive map. Which libraries would be most suitable?

Options :

6406534158696. ✓ Geopandas

6406534158697. ✓ Folium

6406534158698. ✗ Matplotlib

6406534158699. ✗ Shapely

Sub-Section Number :

5

Sub-Section Id :

640653189724

Question Shuffling Allowed :

No

Question Id : 6406531230868 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (341 to 343)

Question Label : Comprehension

Scenario 1:

Sarah is working on a research paper about renewable energy technologies and decides to use an LLM to help her draft some sections.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 341 Question Id : 6406531230869 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Sarah prompts the LLM: "Write a comprehensive overview of solar energy advancements in the last decade."

The instructor wants to ensure that the LLM provides concise, clear, and helpful answers. What is the role of the system prompt in guiding the LLM's behavior?

Options :

6406534158720. ✓ It defines the LLM's tone, behavior, and response style to align with the instructor's teaching goals.

6406534158721. ✗ It provides a list of pre-written responses to be used in all classroom scenarios.

6406534158722. ✗ It prevents the LLM from ever deviating from pre-determined responses, regardless of the context.

6406534158723. ✗ It configures the LLM to only respond to specific commands, disregarding any other input.

Question Number : 342 Question Id : 6406531230870 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

When Sarah submits increasingly complex prompts about solar energy, which of the following statements are TRUE regarding LLM cost and usage

Options :

- 6406534158724. ✓ LLM costs are primarily calculated based on the number of tokens processed
- 6406534158725. ✗ LLM usage is always free for academic research
- 6406534158726. ✓ The total length of both input prompt and output response impacts the overall cost
- 6406534158727. ✗ LLM costs are fixed regardless of the complexity of the query
- 6406534158728. ✗ Token count has no relation to computational resources required

Question Number : 343 Question Id : 6406531230871 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Sarah wants to improve the specificity of her LLM prompt. Which prompt is likely to generate the most precise and useful response?

Options :

- 6406534158729. ✗ "Tell me about solar energy"
- 6406534158730. ✗ "Discuss solar energy advancements"
- 6406534158731. ✓ "Outline 4 key solar energy technological breakthroughs from 2014-2024, including specific efficiency improvements and implementation challenges"
- 6406534158732. ✗ "Write a paragraph about solar energy"

Sub-Section Number :	6
Sub-Section Id :	640653189725
Question Shuffling Allowed :	No

Question Id : 6406531230872 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (344 to 348)

Question Label : Comprehension

Scenario 2

Wildfire Emergency Evacuation Planning

Background:

A wildfire is threatening a rural mountain region, and emergency services need to coordinate evacuation routes for four remote communities. The Emergency Management Center is located at a central command post, and must plan the most efficient evacuation route to ensure rapid and safe community evacuation.

The four communities are:

Pine Ridge Settlement (Latitude: 38.7456, Longitude: -120.8765)
Mountain View Village (Latitude: 38.6987, Longitude: -120.9234)
River Valley Community (Latitude: 38.7123, Longitude: -120.7654)
Forest Creek Hamlet (Latitude: 38.6789, Longitude: -120.8901)

Central Command Post Location:
(Latitude: 38.7200, Longitude: -120.8500)

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 344 Question Id : 6406531230873 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

During the wildfire emergency, multiple teams are updating evacuation plans. Which Git command would a team member use to get the latest evacuation route updates?

Options :

- 6406534158733. ✖ git commit
- 6406534158734. ✖ git checkout
- 6406534158735. ✔ git pull
- 6406534158736. ✖ git add

Question Number : 345 Question Id : 6406531230874 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

An emergency responder needs to search through evacuation logs to find all entries related to "Pine Ridge". Which command would be most useful?

Options :

- 6406534158737. ✖ `find -name "Pine Ridge"`
- 6406534158738. ✔ `grep "Pine Ridge" evacuation-logs.txt`
- 6406534158739. ✖ `ls "Pine Ridge"`
- 6406534158740. ✖ `cat evacuation-logs.txt`

Question Number : 346 Question Id : 6406531230875 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In a resource-constrained emergency environment, which deployment method would be MOST appropriate for the Python-based evacuation management system?

Options :

- 6406534158741. ✖ Running scripts locally on a single emergency responder's laptop
- 6406534158742. ✖ Distributing Python source code to all emergency personnel
- 6406534158743. ✔ Containerized deployment with Docker for consistent execution across multiple devices
- 6406534158744. ✖ Full cloud migration requiring constant internet connectivity

Question Number : 347 Question Id : 6406531230876 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which computational tool would be MOST APPROPRIATE for optimizing the complex evacuation route considering multiple communities?

Options :

- 6406534158745. ✖ Microsoft Excel for manual distance calculations
- 6406534158746. ✖ Google Maps for route visualization
- 6406534158747. ✔ Python libraries like NetworkX or OR-Tools for route optimization
- 6406534158748. ✖ QGIS for static route mapping

Question Number : 348 Question Id : 6406531230877 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What is the primary advantage of using the Haversine formula in emergency route planning?

Options :

- 6406534158749. ✖ Calculating exact road distances
- 6406534158750. ✖ Determining travel time under specific conditions
- 6406534158751. ✔ Computing great-circle distances on a spherical Earth
- 6406534158752. ✖ Measuring elevation changes in mountainous terrain

Sub-Section Number :

7

Sub-Section Id :

640653189726

Question Shuffling Allowed :

No

Question Id : 6406531230878 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (349 to 359)

Question Label : Comprehension

Scenario 3

You are provided with a Gzipped Apache log file from a website having the below data in each row.

- IP: The IP address of the visitor
- Remote logname: The remote logname of the visitor. Typically "-"
- Remote user: The remote user of the visitor. Typically "-"
- Time: The time of the visit. E.g. [01/May/2024:00:00:00 +0000].
- Request: The request made by the visitor. E.g. GET /blog/ HTTP/1.1. It has 3 space-separated parts, namely (a) Method: The HTTP method. E.g. GET (b) URL: The URL visited. E.g. /blog/ (c) Protocol: The HTTP protocol. E.g. HTTP/1.1
- Status: The HTTP status code. If $200 \leq \text{Status} < 300$ it is a successful request
- Size: The size of the response in bytes. E.g. 1234
- Referrer: The referer URL. E.g. https://s-anand.net/
- User agent: The browser used. This will contain spaces and might have escaped quotes.
- Vhost: The virtual host. E.g. s-anand.net
- Server: The IP address of the server.

The fields are separated by spaces and quoted by double quotes ("). Unlike CSV files, quoted fields are escaped via " and not \. All data is in the GMT-0500 timezone and the questions were based in this same time zone.

Use the information given and answer the sub-questions.

Note: Assume that you are using Python codes for log analysis

Sub questions

Question Number : 349 Question Id : 6406531230879 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is a valid log entry based on the provided format?

Options :

6406534158753. ✖ 192.168.1.1 - - [12/Dec/2024:14:05:59 -0500] "GET /image.jpg HTTP/1.1" 200 1234 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/87.0.4280.88 Safari/537.36" www.s-anand.net 192.254.190.217

6406534158754. ✖ 192.168.1.1 - - [12/Dec/2024:14:05:59 -0500] "POST /image.jpg INVALID" 200 1234 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/87.0.4280.88 Safari/537.36" www.s-anand.net 192.254.190.217

6406534158755. ✖ 192.168.1.1 - - [12/Dec/2024:14:05:59 -0500] "PUT /image.jpg HTTP/1.1" OK 1234 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/87.0.4280.88 Safari/537.36" www.s-anand.net 192.254.190.217

6406534158756. ✔ 203.0.113.7 - - [14/Dec/2024:16:45:11 -0500] "GET /index.html HTTP/1.1" 200 3500 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/90.0.4430.93 Safari/537.36" www.s-anand.net 192.254.190.219

Question Number : 350 Question Id : 6406531230880 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following fields are necessary to filter "POST requests made for pages under /images/ from 15:00 to 18:00 on Mondays"?

Options :

6406534158757. ✓ Time, Request, Method, URL

6406534158758. ✗ Time, Method, Status, Size

6406534158759. ✗ Time, Method, URL, Referer

6406534158760. ✗ Time, Status, URL, Server

Question Number : 351 Question Id : 6406531230881 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the time in the log is stored in the format [01/May/2024:00:00:00 -0500], which of the following time formats should you use to parse it correctly?

Options :

6406534158761. ✓ %d/%b/%Y:%H:%M:%S %z

6406534158762. ✗ %Y/%m/%d %H:%M:%S

6406534158763. ✗ %d-%m-%Y %H:%M:%S

6406534158764. ✗ %Y-%m-%dT%H:%M:%SZ

Question Number : 352 Question Id : 6406531230882 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following methods is best for identifying if a log entry's timestamp corresponds to a Sunday?

Options :

6406534158765. ✗ Check if timestamp.dayname() == 'Sunday'

6406534158766. ✗ Check if timestamp.weekday() == 5

6406534158767. ✗ Check if timestamp.strftime('%A') == 'Sunday'

6406534158768. ✓ Check if timestamp.weekday() == 6

Question Number : 353 Question Id : 6406531230883 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How would you identify log entries where the time is between 16:00 and 18:59 on Sundays?

Options :

6406534158769. ✗ Check if 16 <= timestamp.hour < 18

6406534158770. ✓ Check if `16 <= timestamp.hour < 19`

6406534158771. ✗ Check if `16 < timestamp.hour < 19`

6406534158772. ✗ Check if `timestamp.hour in range(16, 20)`

Question Number : 354 Question Id : 6406531230884 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How would you extract the HTTP protocol version (e.g., HTTP/1.1) from the Request field "GET /blog/ HTTP/1.1"?

Options :

6406534158773. ✗ Use `request.split('/')[1]`

6406534158774. ✓ Use `request.split(' ')[2]`

6406534158775. ✗ Use `request.split(' ')[1]`

6406534158776. ✗ Use `request.split('/')[2]`

Question Number : 355 Question Id : 6406531230885 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How would you extract the URL part from the Request field `GET/tamilmp3/HTTP/1.1`?
(Assume that URL is extracted to the variable `url`)

Options :

6406534158777. ✓ Use `request.split(' ')[1]`

6406534158778. ✗ Use `request.split(' ')[0]`

6406534158779. ✗ Use `request.split('/') [1]`

6406534158780. ✗ Use `request.split(' ')[2]`

Question Number : 356 Question Id : 6406531230886 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Based on the answer given to the Previous subquestion, which of the following checks can be used to identify if a request URL is for pages under `/tamilmp3/`?

Options :

6406534158781. ✖ `'/tamilmp3/' in url`

6406534158782. ✔ `url.startswith('/tamilmp3/')`

6406534158783. ✖ `url == '/tamilmp3/'`

6406534158784. ✖ `url.endswith('/tamilmp3/')`

Question Number : 357 Question Id : 6406531230887 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How would you identify if the log entry's HTTP status code indicates a redirection request?
(Assume that status is correctly stored in the variable 'status')

Options :

6406534158785. ✖ `status in [301, 302, 303]`

6406534158786. ✖ `200 < status < 400`

6406534158787. ✔ `300 <= status < 400`

6406534158788. ✖ `status in range(299, 400)`

Question Number : 358 Question Id : 6406531230888 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

After filtering the log entries (in a list `filtered_entries`), which of the following is the best way to count the number of requests with a 404 status code for pages under `/error/`?

Options :

6406534158789. ✖ `Use len(filtered_entries)`

6406534158790. ✖ `Use len(entries)`

6406534158791. ✖ `Use sum(1 for entry in entries)`

6406534158792. ✔ `None of these`

Question Number : 359 Question Id : 6406531230889 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the count of log entries appears much larger than expected, which of the following could be the reason?

Options :

6406534158793. ✖ `The filter for Monday is incorrect.`

6406534158794. ✖ `The filter for time range is incorrect.`

6406534158795. ✖ `The filter for HTTP method is incorrect.`

6406534158796. ✔ Any of these could be the reason.

Sub-Section Number : 8
Sub-Section Id : 640653189727
Question Shuffling Allowed : No

Question Id : 6406531230890 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (360 to 363)

Question Label : Comprehension

Scenario 4

A climate science professor is developing an AI-powered learning assistant called ClimateBot to help students understand complex climate modeling techniques and computational approaches to environmental data analysis.

Guidelines for ClimateBot:

- Engage students through thought-provoking questions
- Encourage independent problem-solving
- Guide students to discover solutions autonomously
- Provide conceptual insights, not direct answers
- Help students develop robust computational and analytical skills

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 360 Question Id : 6406531230891 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What is the primary purpose of the system prompt in controlling an AI learning assistant's behavior?

Options :

6406534158797. ✖ To create a fixed set of predetermined responses for every possible student query

6406534158798. ✔ To define the core communication strategy, learning approach, and interaction guidelines

6406534158799. ✖ To replace the need for human instructor guidance completely

6406534158800. ✖ To limit the AI's ability to understand complex scientific concepts

Question Number : 361 Question Id : 6406531230892 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Choose the **best system prompt** that achieves the instructor's goals.

Options :

6406534158801. ✖ "You are a climate science assistant. Provide direct answers and complete code solutions whenever students ask."

6406534158802. ✔ "You are an interactive learning assistant for climate science. Guide students through complex concepts by asking reflective questions. Avoid giving direct solutions. Encourage independent thinking and help students develop problem-solving skills."

6406534158803. ✖ "You are a simple computational tool. Provide exact numerical answers and full code implementations for any climate modeling question."

6406534158804. ✖ "You are a climate science assistant. Help students and tell them to consult textbooks for all information."

Question Number : 362 Question Id : 6406531230893 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A student asks: "How do I calculate the global warming potential of different greenhouse gases?"

Choose the most pedagogically effective response:

Options :

6406534158805. ✖ "The global warming potential (GWP) is calculated by multiplying the gas's radiative efficiency by its atmospheric lifetime. Here's the exact formula: [provides complete formula and calculation method]"

6406534158806. ✖ "Global warming potential (GWP) is determined by factors like a gas's molecular structure, atmospheric lifetime, and interaction with radiation. Different gases have different GWPs based on these characteristics."

6406534158807. ✔ "Let's break this down. What do you already know about greenhouse gases and their impact on climate? Have you considered how different molecules might have varying abilities to trap heat?"

6406534158808. ✖ "I cannot help you with this calculation. Please refer to your textbook for a complete explanation."

Question Number : 363 Question Id : 6406531230894 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Assume that the ClimateBot, an advanced learning assistant for climate science students. Its primary mission is to foster critical thinking, scientific reasoning, and deep understanding of complex environmental systems by using a **RAG (Retrieval-Augmented Generation) system**. The RAG system works as follows:

1. **Climate Data Storage:** Comprehensive climate research materials (like reports, research papers and notes) are split into **smaller chunks**. Each chunk is vectorized using an embedding model.

2. **Vector Database:** The vectorized chunks are stored in a **FAISS or Weaviate vector database**.
3. **Retriever:** When a student asks a question, the system retrieves the most relevant content (based on semantic similarity) from the vector database.
4. **LLM Contextualization:** The retrieved content is added as context in the assistant's prompt, which is then used by the **LLM (like GPT-4)** to generate a personalized, context-aware response.

Which of the following **correctly describes the process flow of a RAG-based system** for the IITM TDS Teaching Assistant?

Options :

Student Query → LLM Direct Answer

6406534158809. ✖ → Response Sent Back

Student Query → Chunk Course Material

→ Vectorize Query → Retrieve Matching

6406534158810. ✖ Content → LLM Generates Response

Student Query → Vectorize Query

→ Retrieve Content from Vector

Database → Pass Retrieved Content to LLM

6406534158811. ✔ → LLM Generates Context-Aware Response

Student Query → Use Rule-Based

System to Find Answer → Use Pre-Written

6406534158812. ✖ Response → Response Sent Back